



MUHAMMAD HASEEB

Research Scholar & Physicist

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<https://mhaseebbwp.github.io/resume/>

SKILLS

- Expertise in **Wien2k**, **OLCAO**, **Abinit** for first-principles calculations
- Proficient in **VELAS** and **EIA Tools** for mechanical anisotropy analysis
- Skilled in **Origin**, **Gnuplot**, and **XMGrace** for graph plotting and data analysis
- Experienced with **Crystal Maker**, **Vesta**, **Meshlab**, **View3Dscene** for material visualization
- Strong programming skills in **C**, **C++**, **Python**
- Proficient in **MATLAB** for data processing and simulation
- Advanced use of **MS Office** and related applications

LANGUAGES

- ENGLISH
- URDU
- PUNJABI

RESEARCH INTEREST

- Thermomagnetic and thermoelectric material design for energy applications
- First-principles modeling of semiconductors and advanced functional materials
- Spintronic materials and spin-based magnetic phenomena
- Computational study of novel materials for optical and electronic applications
- Anisotropic mechanical and phononic properties of advanced materials

WORK EXPERIENCE

Lecturer Physics May 2024 - Present
Moon System of Education I Bahawalpur, Pakistan

Dedicated to fostering a deep understanding of physics through engaging teaching and innovative methods.

Visiting Lecturer Feb 2023 - June 2023
 Institute of Physics (IUB) I Bahawalpur, Pakistan

Teaching Asistant Sep 2022 - Jan 2023
 Institute of Physics (IUB) I Bahawalpur, Pakistan

Visiting Lecturer Mar 2021 - July 2022
 Institute of Physics (IUB) I Bahawalpur, Pakistan

EDUCATION

PhD Physics 2022 - Present
Institute of Physics (IUB) I Bahawalpur, Pakistan

Density Functional Insights of Oxychalcogenide Materials for Energy Harvesting Applications under the supervision of **Dr. Altaf Hussain (Professor)** in Materials Simulation Group (MSG).

M. Phil. Physics

Institute of Physics, The Islamia University of Bahawalpur (IUB) I Bahawalpur, Pakistan

DFT based simulation is used to study structural, electronic, magnetic and optical properties of carbides under the supervision of **Dr. Altaf Hussain** (Associate Professor) in Institute of Physics (IoP). The topic of my research is “Mechanical, Thermal, Optical and Electronic Properties of Nanolaminated Ternary Carbide $U_3Al_3C_5$ ” Obtain **3.93 CGPA out of 4.0** in thesis research project. I have also published a research paper during my MPhil research work in the “**Journal of Materials Research and Technology**” entitled “Structural stability, electronic structure, mechanical and optical properties of MAX phase ternary Mo_2Ga_2C , Mo_2GaC and Mo_3GaC_2 carbides”.

2015 - 2019

BS (Honors) Physics

Institute of Physics, The Islamia University of Bahawalpur (IUB) I Bahawalpur, Pakistan

Solid state physics, laser, nuclear physics, quantum mechanics and electrodynamics as majors, obtained **3.88 CGPA out of 4.0** with **3rd position** in 6th Semester, **B+ grade** in review research project, and report titled “Density Functional Theory based Electronic Structure Studies of Cr-Al-C Max System”.

RESEARCH PUBLICATIONS

2025 | Research Publication in International Journal

- **Journal:** Inorganic Chemistry Communications - Elsevier
- **Impact Factor:** 4.4
- **Journal Rank:** Q1
- **Title of Publication:** Comprehensive investigation of the physical properties of X_5SiC_2 ($X = Cr, Mo, W$) silicides: A first-principles approach for high-temperature applications [🔗](#)
- **Contribution:** Corresponding Author
- **Role:** Writing - review & editing, Project administration.

2025 | Research Publication in International Journal

- **Journal:** Materials Science in Semiconductor Processing - Elsevier
- **Impact Factor:** 4.2
- **Journal Rank:** Q1
- **Title of Publication:** Understanding the properties of novel ternary $LiXN_2$ ($X = V, Nb$) nitrides for energy harvesting applications: A first-principles study [🔗](#)
- **Contribution:** Corresponding Author
- **Role:** Writing - review & editing, Project administration.

2024 | Research Publication in International Journal

- **Journal:** Solid State Communications - Elsevier
- **Impact Factor:** 2.1
- **Journal Rank:** Q2
- **Title of Publication:** Properties of bismuth based Bi_2A_3 ($A = S, Se, Te$) chalcogenides for optoelectronic and thermoelectric applications [🔗](#)
- **Contribution:** 2nd Author
- **Role:** Writing - review & editing, Project administration.

2024 | Research Publication in International Journal

- **Journal:** Materials Today Communications - Elsevier
- **Impact Factor:** 3.7
- **Journal Rank:** Q2
- **Title of Publication:** First-principles calculations of $ScMC_2$ ($M = Fe, Co, Ni, Cu$) ternary carbides: A suitable candidate for shielding purpose [🔗](#)
- **Contribution:** 4th Author
- **Role:** Writing – review & editing.

2024 | Research Publication in International Journal

- **Journal:** Physica B: Condensed Matter - Elsevier
- **Impact Factor:** 2.8
- **Journal Rank:** Q2
- **Title of Publication:** First-principles calculations on electronic, mechanical and optical properties of ScMC₂ (M = Ti, V, Cr, and Mn) ternary carbides [🔗](#)
- **Contribution:** 3rd Author
- **Role:** Formal analysis, Methodology.

2024 | Research Publication in International Journal

- **Journal:** Materials Science in Semiconductor Processing - Elsevier
- **Impact Factor:** 4.2
- **Journal Rank:** Q1
- **Title of Publication:** First-principles study of structural, electronic, mechanical, optical, thermodynamic and thermoelectric properties of ternary ZnSnN₂ and ZnMoN₂ nitrides [🔗](#)
- **Contribution:** Corresponding Author
- **Role:** Writing - review & editing, Project administration.

2024 | Research Publication in International Journal

- **Journal:** Journal of Physics and Chemistry of Solids - Elsevier
- **Impact Factor:** 4.3
- **Journal Rank:** Q2
- **Title of Publication:** First-principles investigation of physical properties of transition metal-based ternary TM₅Si₃C (TM = Nb, Mo, Ta, W) silicides of Nowotny phase [🔗](#)
- **Contribution:** Corresponding Author
- **Role:** Writing - review & editing, Project administration.

2023 | Research Publication in International Journal

- **Journal:** Materials Today Communications - Elsevier
- **Impact Factor:** 3.7
- **Journal Rank:** Q2
- **Title of Publication:** Ab initio study of electronic structure, bonding and optical properties of ternary A₂B₂In (A = Ca, Eu and B = Pd, Pt) intermetallics [🔗](#)
- **Contribution:** 2nd Author
- **Role:** Software, Validation.

2023 | Research Publication in International Journal

- **Journal:** Vacuum - Elsevier
- **Impact Factor:** 3.8
- **Journal Rank:** Q2
- **Title of Publication:** First-principles study on physical properties of actinide-based ternary (UC)_mAl₃C₂ (m = 1, 2, 3) carbides [🔗](#)
- **Contribution:** 2nd Author
- **Role:** Writing - review & editing, Project administration.

2022 | Research Publication in International Journal

- **Journal:** Journal of Solid-State Chemistry - Elsevier
- **Impact Factor:** 3.2
- **Journal Rank:** Q2
- **Title of Publication:** DFT based comparative study of physical properties of Y₃AlC₃, YAl₃C₃ and Y₃AlC carbides [🔗](#)
- **Contribution:** 4th Author
- **Role:** Software, Validation.

2021 | Research Publication in International Journal

- **Journal:** Journal of Materials Research and Technology - Elsevier
- **Impact Factor:** 6.2
- **Journal Rank:** Q1
- **Title of Publication:** Structural stability, electronic structure, mechanical and optical properties of MAX phase ternary $\text{Mo}_2\text{Ga}_2\text{C}$, Mo_2GaC and Mo_3GaC_2 carbides [🔗](#)
- **Contribution:** 2nd Author
- **Role:** Writing - review & editing, Project administration.

HONOURS AND AWARDS

15 November 2018 | PM's Merit Laptop Award by Government of Punjab, Pakistan

Prime Minister's Laptop scheme | Punjab, Pakistan [🔗](#)

Representative student of Department of Physics receives the laptop award in distribution ceremony chaired by Provincial Minister of Punjab.

13 December 2014 | CM's Merit Laptop Award by Government of Punjab, Pakistan

Chief Minister's E-Youth Initiative | Punjab, Pakistan [🔗](#)

Receive the laptop award in distribution ceremony chaired by Provincial Minister of Punjab.

31 December 2013 | CM's Ujala Program – Solar Home systems for Students

Chief Minister's Ujala Program | Punjab, Pakistan [🔗](#)

Receive solar panels by excellent performance.

WORKSHOPS, SEMINARS, CONFERENCES AND TRAININGS

November 2024 | Certification of Reviewing by Vacuum - Elsevier Journal

Vacuum - Elsevier Journal [🔗](#)

Awarded by the Vacuum Journal for reviewing four papers between January and November 2024, in recognition of valuable contributions to the journal.

September 2023 | Certification of Participation/Oral Presentation

International Conference on Materials Science and Nanotechnology MSNANO-23 Faisalabad | Pakistan [🔗](#)

Talk given at “6th International Conference on Materials Science and Nanotechnology MSNANO-23” scheduled on 25-27 September 2023 at Quaid e Azam Auditorium New Campus Government College University Faisalabad.

July 2023 | Certification of Participation

Three Days workshop on Quantum Chemistry Thermodynamic approach organized by Zastra Innvations Pvt Ltd [🔗](#)

Participated in the three days' workshop titled “Fast and Robust Quantum Chemistry” and “Solvation Chemistry for modelling and prediction of fluid phase properties” conducted by the Zastra Innvations Pvt Ltd from 25th to 27th July, 2023.

July 2023 | Certificate of 10 Days Training on Online Teaching Tools

Edexmy LMS Pakistan first public LMS | Pakistan [🔗](#)

Ten Days Training Offered by Edexmy Tuition Hub, Edexmy LMS Pakistan first public LMS | Pakistan

April 2023 | Certification of Participation

Five Days FDP (Faculty Development Program) at Vellore Institute of Technology, Chennai | India [🔗](#)

Participated in the five days FDP titled “Modelling, Simulation & Fabrication of Advanced Electronic Devices” conducted by the School of Electronics Engineering (SENSE) from 15th to 19th April, 2023 at Vellore Institute of Technology, Chennai, India.

December 2022 | Certificate of Participation/ Training

Training Offered by Directorate of Teaching Assistants & Research Assistants, IUB | Pakistan

One Day Training Offered by Directorate of Teaching Assistants & Research Assistants, The Islamia University of Bahawalpur, Pakistan.

August 2022 | 1 st Pedagogical Training 2022 Certificate

Training Organized by the Department of Education, IUB | Pakistan

Offered by The Islamia University of Bahawalpur, Pakistan

February 2022 | Certification of Participation/Oral Presentation

International Conference on Solid State Physics (ICSSP) Lahore | Pakistan

Talk given at “International Conference on Solid State Physics” organized by Center of Excellence Solid State Physics in Punjab University Lahore.

January 2021 | Python for Everybody Specialization learning program

Offered by University of Michigan through Coursera Online Learning

Completed the Python for Everybody Specialization, offered by the University of Michigan via Coursera, in January 2021.

March 2019 | Certification of Participation

International School on Physics & Allied Disciplines” National Centre for Physics (NCP), Islamabad | Pakistan

Participate as a volunteer organizer in “International School on Physics & Allied Disciplines” organized by National Centre of Pakistan in Islamabad.

RECOMMENDATIONS

Dr. Saeed Ahmad Buzdar

- Professor
- Vice Chancellor
- Thal University Bhakkar

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Dr. Altaf Hussain

- Professor
- Director Institute of Physics.
- The Islamia University of Bahawalpur (IUB), Bahawalpur, Pakistan

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Dr. Muhammad Amir Rafiq

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- Institute of Physics.
- The Islamia University of Bahawalpur (IUB), Bahawalpur, Pakistan

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